Physics-informed Neural Networks for the phonon Boltzmann Transport Equation (BTE) and Magnetic Resonance Electrical Property Tomography (MR-EPT)

The Boston University photonics center

BOSTON UNIVERSITY

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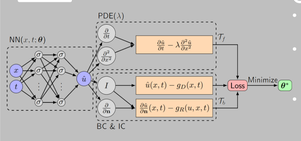
U.S. Army Research Office
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Abstract

Physics-informed neural networks (PINNs) enable us to leverage Deep Neural Network optimizations in physics and engineering applications

- Automatic differentiation enables PINNs to embed PDEs into their loss functions with analytical derivatives
- Training with randomly assigned collocation points (mesh-free), satisfying PDE on the entire domain
- Developed novel architecture and loss function, leading to increased contrast and precision in EPT problem
- Hard constraints enable efficient forward BTE modeling

PINNs



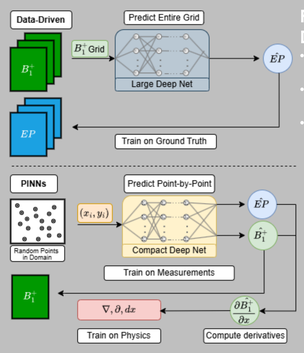
Lu Lu et al., SIAM Review , 63, 1, (2020)

Goals of PINN

- Minimize the residual form of a PDE
- Minimize Boundary (BC) or Initial Conditions (IC) for forward problems
- Minimize total loss on randomly sampled collocation points inside the domain and on the boundary

Properties of PINN

- Accurate mesh-free method
- Leverages automatic network differentiation for analytical derivatives
- Ideal framework for solving complex inverse problems at minimal computational cost
- Does not require large datasets – Not a data-driven ML technique



Properties of Data-Driven

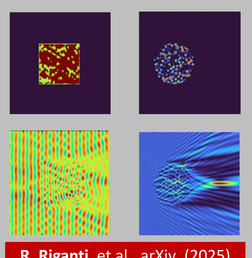
- Large models with long training times
- Dependence on large dataset
- No consideration of underlying physics can lead to unphysical results

Additional Properties of PINN

- Require retraining for new BC/IC configurations
- Quantifiable physics error
- Compact models

Prof. Dal Negro's Nano Research Group

Recent Paper on Multiscale for Photonic Inverse Design



R. Riganti, et al., arXiv, (2025)

Highly Successful PINN Papers

R. Riganti and L. Dal Negro, Appl. Phys. Lett., 123, 171104 (2023)

Y. Chen, et al., APL Photonics, 7, 010802 (2022)

Y. Chen, et al., Opt. Express, 28, 11618–11633 (2020)

Developed PINN for MR-EPT

Helmholtz

$$\nabla^2 B_1^+ + k_0^2 \varepsilon_c B_1^+ = \left(\frac{\partial B_1^+}{\partial x} - i\frac{\partial B_1^+}{\partial y}\right)(g_x + ig_y) + \frac{\partial B_1^+}{\partial z} g_z$$

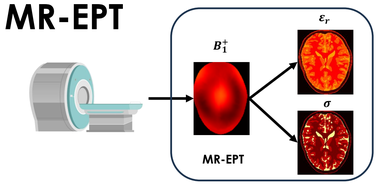
Where $g = \nabla \ln \varepsilon_c$

Magnetic Field: B_1^+

Electrical Properties(EP): $\varepsilon_c = \varepsilon_r - \frac{i\sigma}{\omega \varepsilon_0}$

Relative Permittivity: ε_r

Conductivity: σ



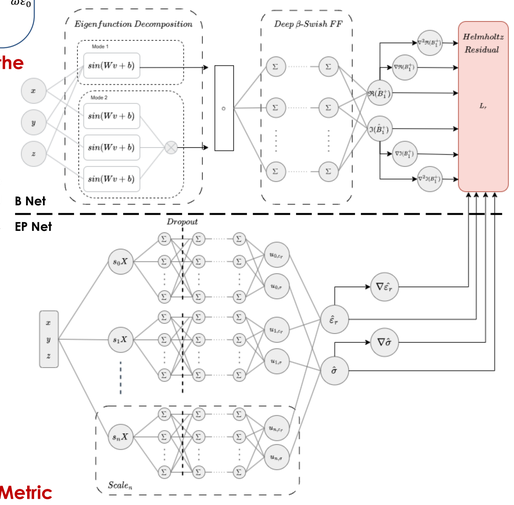
Our Developed Model for the Magnetic Field (B) and Electrical Properties (EP) Networks

Developed Eigenfunction Decomposition PINN (EF-PINN)

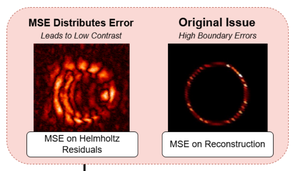
- Uses eigenmodes of the Helmholtz as a preprocessing step
- Provides a functional basis for the PINN
- Allows for frequency tunning

Developed Multiscale Dropout PINN (MS-DO-PINN)

- Utilizes multiple sub-networks called "scales" for high-precision reconstruction
- Dropout layers randomly drop neurons during training
- Dropout reduces overfitting and forces network to explore nearby solutions



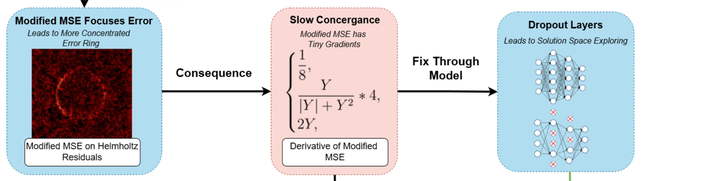
Our Proposed Modified Mean Squared Error (MSE) Metric



Achieving High-Contrast, High-Precision Reconstruction on Noisy Measurement Data

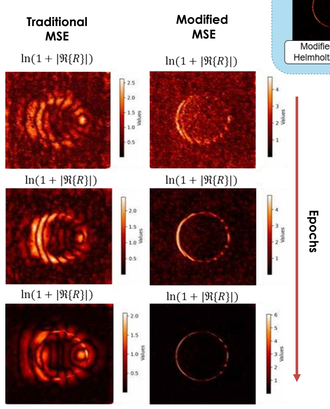
Collaborators from New York University, Department of Radiology

- Developed Physics-Informed Fourier Networks(PIFON) for MR-EPT (2023)
- This works extends and improves PIFON



Comparison Between MSE functions

Logarithmic plots of the Helmholtz residual over training epochs using:

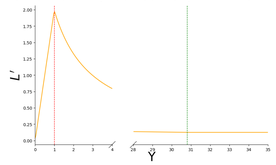


Modified MSE Definition

$$L = \begin{cases} \frac{|X|}{8}, & |X| > 30.8 \\ \log_{1.286}(|X|+1), & 1 < |X| \le 30.8 \\ X^2, & |X| \le 1 \end{cases}$$

$$L' = \begin{cases} \frac{1}{8}, & |X| > 30.8 \\ \frac{X}{|X|+X^2} * 4, & 1 < |X| \le 30.8 \\ 2X, & |X| \le 1 \end{cases}$$

Derivative of L (Modified MSE) with Axis Break



Developed PINN for the phonon BTE

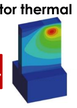
$$f = f_{eq} + f_{neq}$$

$$\mathbf{v} \cdot \nabla f + \frac{f_{neq}(T)}{\tau(T)} = 0$$

$$\nabla \cdot \sum_p \int_0^{\omega_{max}} \int_0^{4\pi} \mathbf{v} \hat{f} d\Omega d\omega = 0$$

BTE for transistor thermal modeling

Y. Hu, et al., Fundamental Research, 4, 907–915 (2024)



Our Developed Models for Feq and Fneq Networks

Integrated Hard Constraints

- Post processing step, done with multiplying the Network output with nullifying function
- Satisfy boundary conditions by construction
- Reduce model and training grid size

Integrated Scaling Initialization

- Leads to faster convergence

Improvement over current State-of-the-art PINN for BTE:

- Training time reduced from 4 hours to 10 minutes

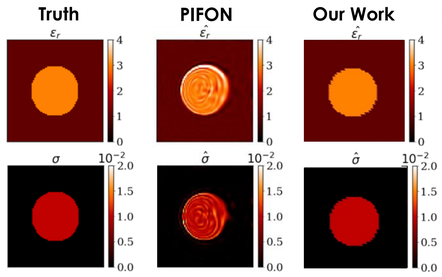
Our Results

EPT Under Signal-to-Noise Ratio of 200

Discrepancy measured with Peak Normalized Absolute Error

2D Cylinder Under Plane Wave

Permittivity: **.005%**
Conductivity: **.009%**



Xinling Yu et al., arXiv:2302.11883, (2023)

MRI-like 2D Cylinder under Wire Source

Permittivity: **.12%**
Conductivity: **.08%**

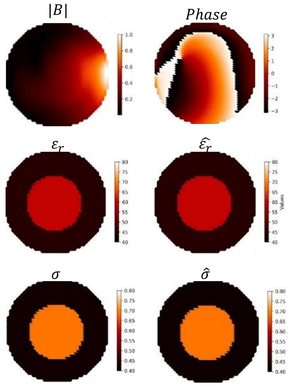
Grey Matter and White Matter EPs

300 MHz (7T)

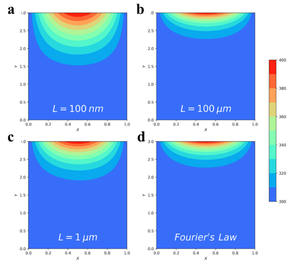
Domain r = .11cm

Maximum point-by-point error of 1%

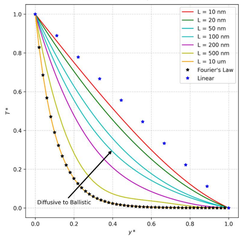
Rounding predicted EPs to the closest material produces 100% accurate reconstruction



2D Silicon with Top Heat Source



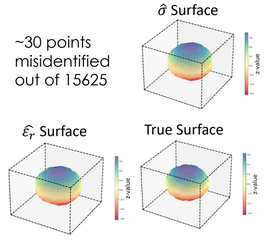
Center axis cut of Temperature Profile



3D Sphere Under Plane Wave

Permittivity: **.12%**
Conductivity: **.17%**

~30 points misidentified out of 15625



Conclusions

Magnetic Resonance Electrical Property Tomography (MR-EPT)

- Developed a frequency-tuning Eigenfunction-PINN (EF-PINN) for modeling noisy magnetic field data from MRI machines
- Developed a flexible and expressive Multiscale Dropout PINN (MS-DO-PINN) for high-precision reconstruction of electrical properties
- Developed a novel Modified MSE loss function to concentrate error along material interfaces, resulting in improved solution accuracy
- Demonstrated an improvement over the PIFON framework (state-of-the-art PINN for MR-EPT) reducing errors from ≤ 10% to ≤ .01%
- Demonstrated high-precision EPT on MRI-like 2D and 3D, achieving errors of ≤ .1%

Phonon Boltzmann Transport Equation (BTE)

- Developed an efficient phonon BTE Hard Constraint PINN (HC-PINN) greatly reducing computational resources
- Demonstrated HC-PINN efficiency for phonon BTE by reducing the computation time from 4 hours to 10 min, compared with the state-of-the-art PINN BTE solver

Future Work

Magnetic Resonance Electrical Property Tomography (MR-EPT)

- Expand results into in-vivo MRI measurements
- Model complex heterogeneous 3D objects in the multiple scattering regime

Phonon Boltzmann Transport Equation (BTE)

- Validate with experimental data and expand into complex multi-material devices